



Medical Microbiology

Third stage- College of Dentistry

Gram Stain

Lab. 4

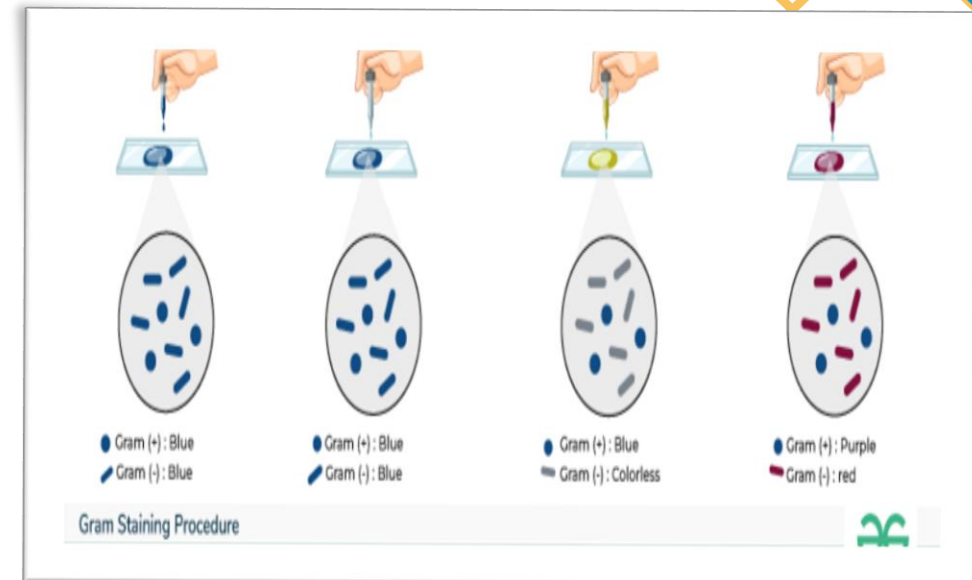
Prepared by

PhD. Dheyaa Naji Hamza

M.Sc. Zainab Majid Mohammed

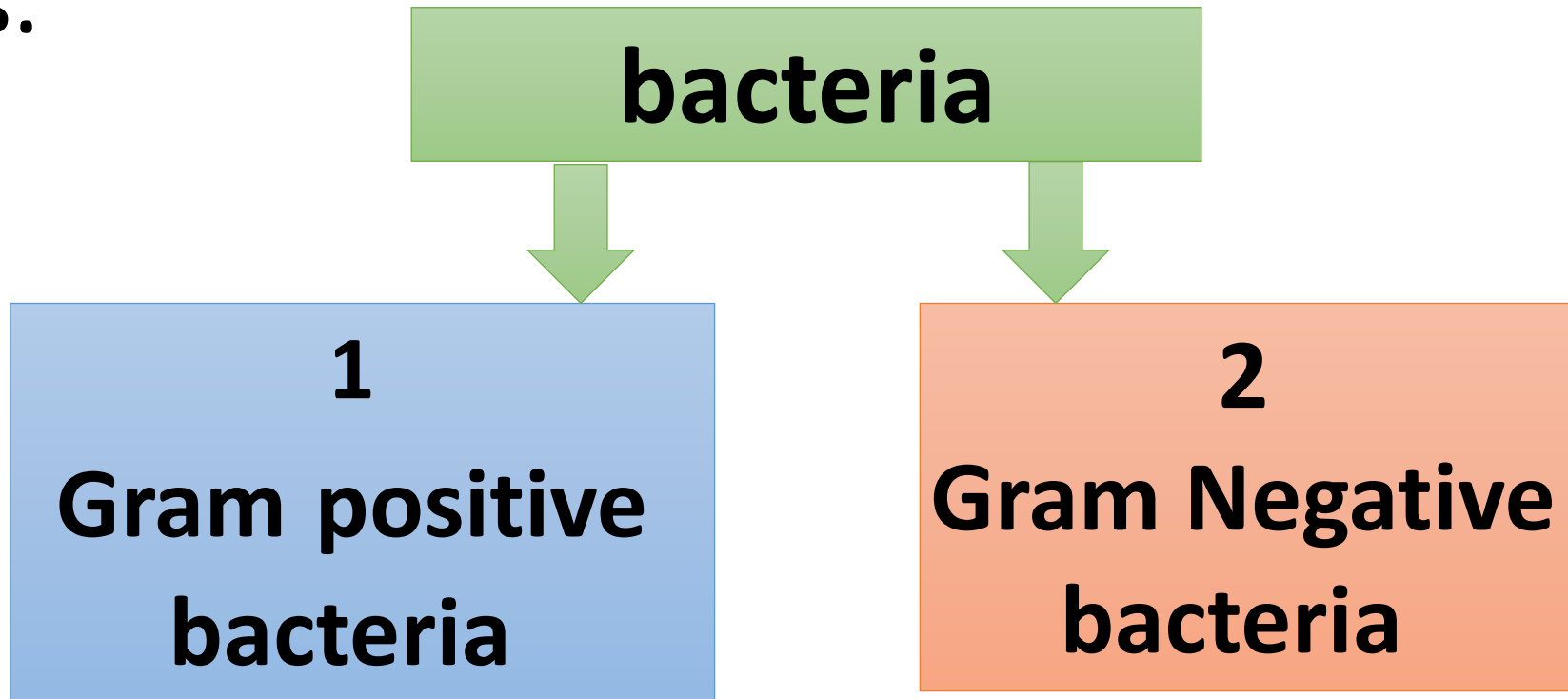
Learning Objectives:

- Knowing gram stain .
- Learning Procedure of Gram Staining.
- Understand the differences between Gram positive and Gram negative.



Grams Stain:

Gram staining: is a method used to differentiate bacteria based on the chemical and physical properties of their cell walls, classifying them into two major groups.

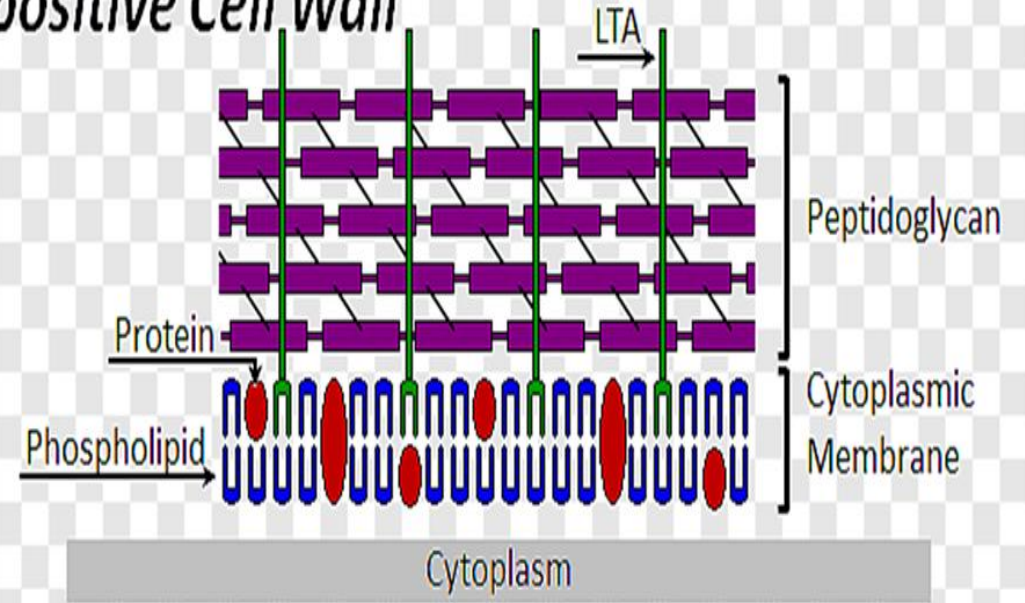


Bacterial cell wall:

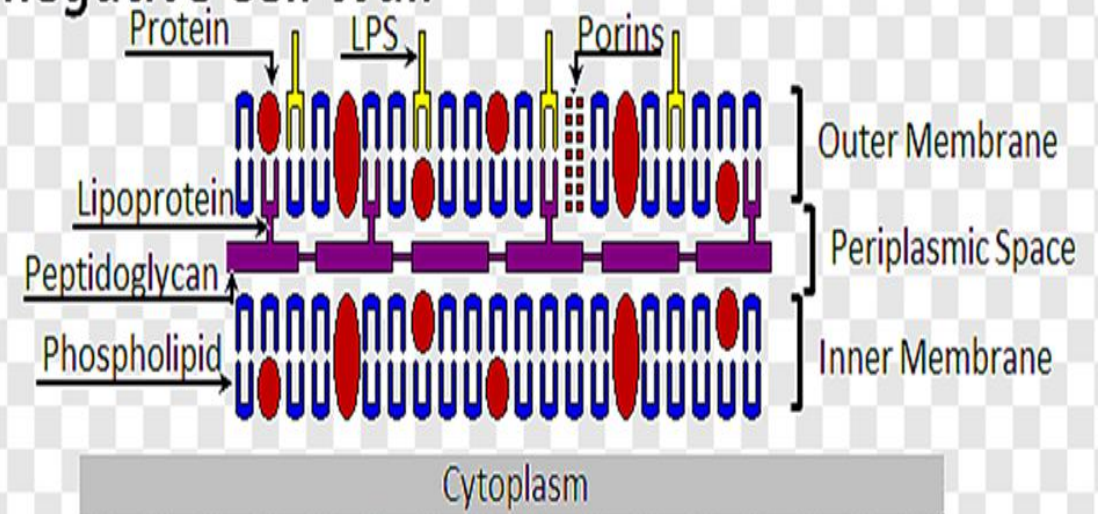
- The cell wall is the outer structure that surrounds the internal components of the cell and the cytoplasmic membrane.

- Its composition differs between Gram-positive and Gram-negative bacteria.

Gram-positive Cell Wall



Gram-negative Cell Wall

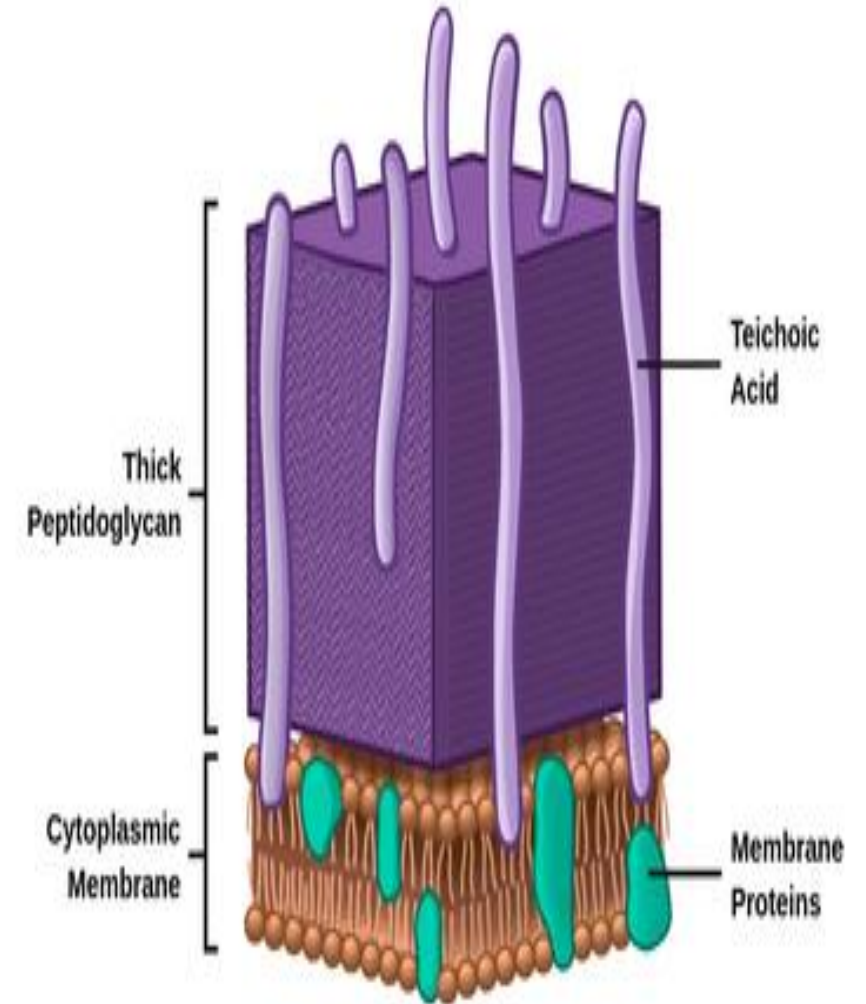


Gram-Positive Bacteria

Gram-positive bacteria have a thick peptidoglycan cell wall.

Peptidoglycan: is a polymer composed of sugars and amino acids that forms a mesh-like layer outside the plasma membrane, giving structure and protection to the bacterial cell.

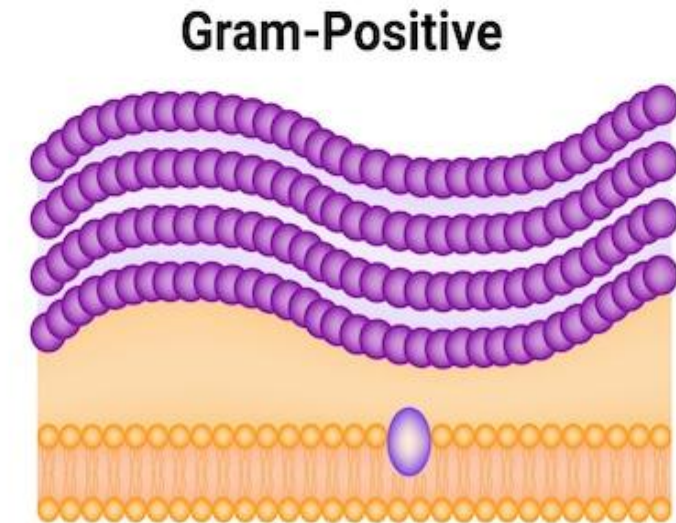
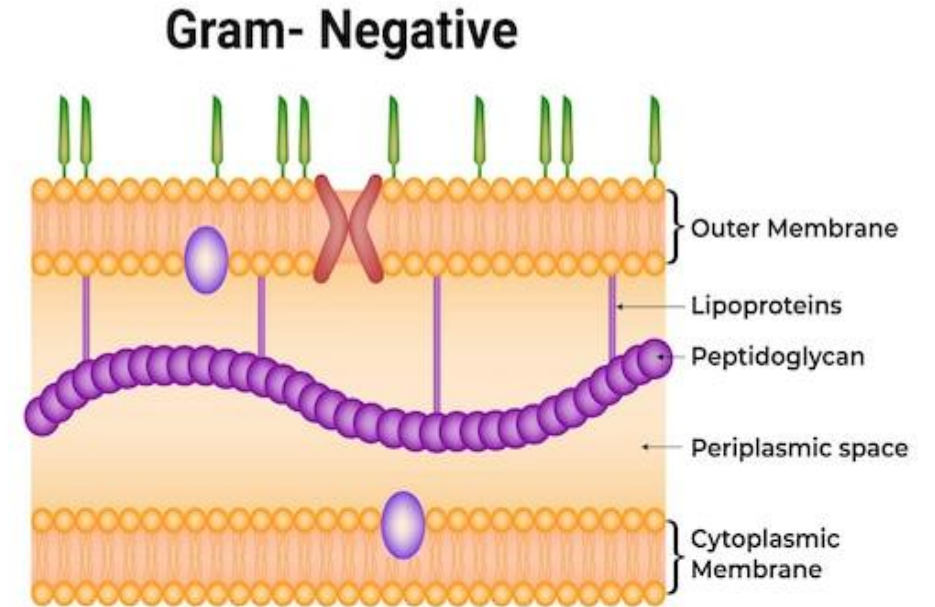
In Gram-positive bacteria, a membrane-bound molecule called teichoic acid is located between the cell wall and the cell membrane.



Gram Negative Bacteria

Gram negative bacteria have an outer membrane of phospholipids and bacterial Lipopolysaccharides outside of their thin peptidoglycan layer.

The space between the peptidoglycan and cytoplasmic membrane is called the periplasmic space.



Principle of Grams staining:

The structure of the **organism cell wall** determine the organism is gram positive or negative.

1- Crystal violet: all bacteria take crystal violet, so appears violet.

2- Iodine: crystal violet – iodine (CV-I) complex is formed.

3- Acetone: bacteria with high lipid content loose complex (appear colourless) but bacteria with less Lipid content retains (CV_I) complex (appear violet).

4- Safranine: only colorless bacteria takes (appear pink).

Gram Staining Procedure:

- 1-Take a clean slide.
- 2-Using a sterile loop, place a small amount of the bacterial suspension onto the slide to prepare a smear.
- 3-Allow the smear to air dry, then heat-fix it by briefly passing the slide through a flame.
- 4-Cover the smear with **crystal violet** stain and let it sit for **1 minutes**. Rinse gently with water.
- 5-Put **iodine solution**, let it stand for **1 minute**, then rinse with water.

6. Then ,wash with 95% **alcohol** or **acetone** for about **10 seconds** and rinse with water.

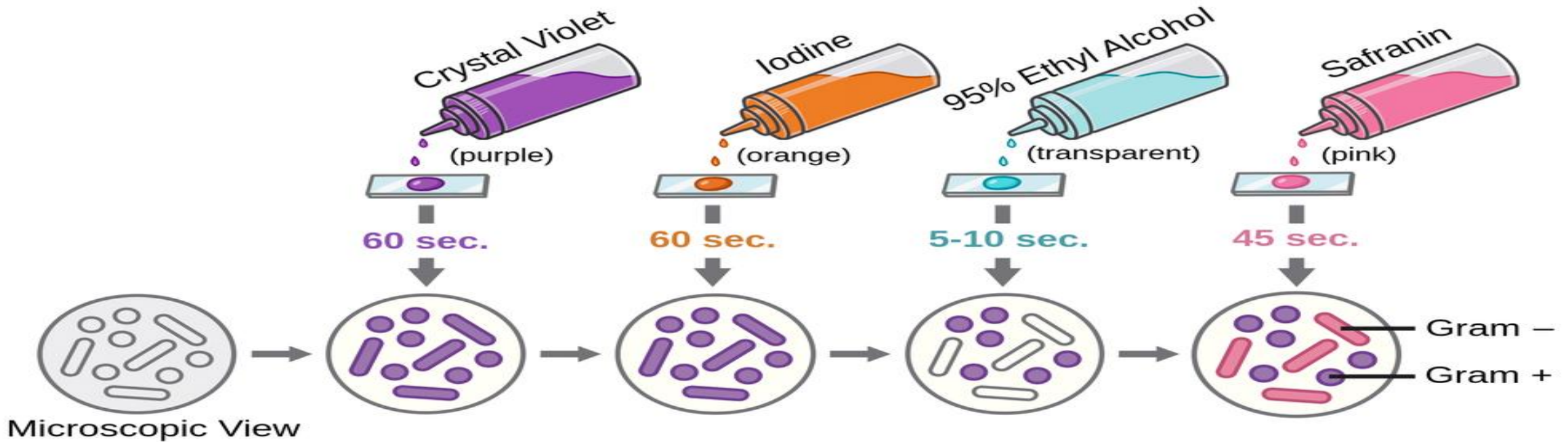
7. Add **safranin** for about **1 minute** and wash with water.

8. Air dry and Observe under Microscope.

Result

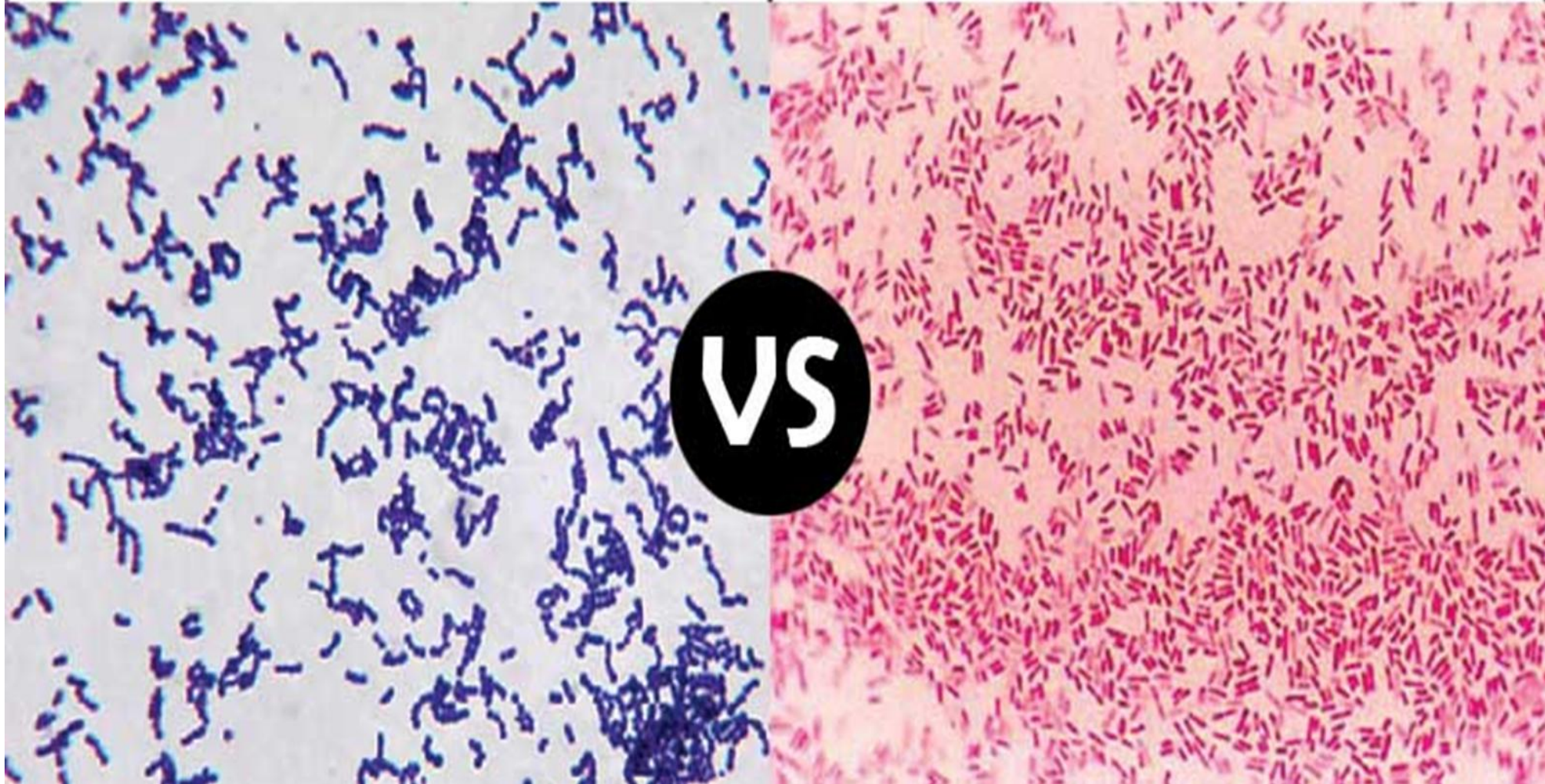
Gram Positive: **Purple Color**

Gram Negative: **Red Color**



Gram Positive Bacteria

Gram Negative Bacteria



Difference between Gram positive and Gram negative bacteria

Gram positive bacteria	Gram negative bacteria
Simple cell wall	More complex cell wall
Thick peptidoglycan cell wall layer	Thin peptidoglycan cell wall layer
No outer lipopolysaccharide wall layer	outer lipopolysaccharide wall layer
Retain crystal violat/iodine	Retain safranin
Appear (Blue/Purple)	Appear (Pink/Red)



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Any Question?



Thank you